Do Now · retrieval

Lesson 075 · Big O: Quadratic, Exponential and Growth

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does O(n) differ from O(1)?

Q2

How many halvings 32→1?

Q3

Which algorithm naturally explores include/exclude branches?

Q4

A rectangular 4×3 zero-indexed array holds how many elements? Give its last valid index pair.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Two nested n loops perform n×n operations, producing O(n²) time when each inner operation is constant.

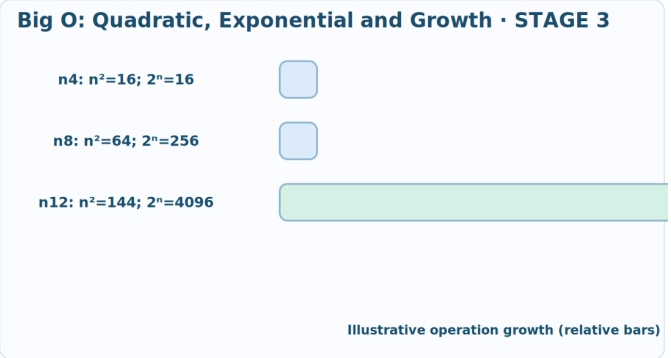
Core idea 2

Explain — A binary decision at each of n positions can create 2^n possibilities. This exponential growth quickly outruns polynomial methods.

Core idea 3

Evaluate — Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

quadratic — Growth proportional to n^2.

exponential — Growth like 2^n.

pruning — Avoid exploring impossible branches.

scalability — Handling larger inputs feasibly.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

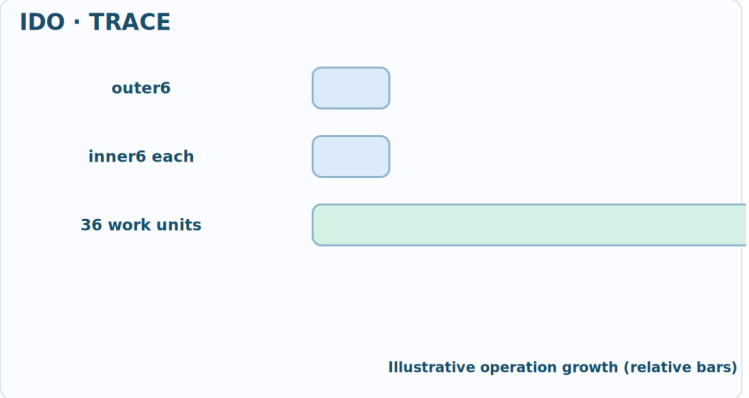
Problem

For n=6, calculate total iterations of two independent nested n loops.

Follow the worked method

1. Explain — Each of the six outer iterations runs six inner iterations.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Every new independent binary choice doubles the count.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For n=5 and n=10, calculate n^2 and 2^n values. [6]

Question 2

Explain time growth of a nested n×n loop and classify. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A problem has 30 binary choices. Evaluate full subset enumeration and one way to reduce exploration. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An O(2^n) approach grows only twice as much when n doubles.

Use key vocabulary from memory

1. Explain quadratic in your own words.

2. Explain exponential in your own words.

3. Explain pruning in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.